Statistical Modelling of Packet-Loss Patterns in Computer Networks: Distribution Fitting, Burstiness Testing and Bayesian Estimation

BAMAT207 Probability and Statistics: Case Study Report

***Abstract*: The loss rate of a network is usually reported as a single number, and the simplest models assume that each packet is lost independently of the others. If that held, the number of losses per second would be Poisson and the gap from one loss to the next would be exponential. In this study we check how much of this survives when losses arrive in bursts. Three networks (wired broadband, campus Wi-Fi and a mobile hotspot) were simulated with a two-state Gilbert-Elliott model, each sending 50 packets per second for 1 hour, which comes to 180,000 packets per network. We fitted models for the counts and for the waiting times, tested whether neighbouring packets are independent, and estimated the loss rates with a Bayesian Beta-Binomial model. Independence did not hold on any of the three networks. The variance of the per-second loss count was 1.59, 5.31, 10.17 times the mean, where a Poisson model gives 1, and after a single loss the next packet was 31 to 43 times more likely to be lost. This shows up directly in the error bars. Over 2,000 simulated captures, the standard 95% interval for the loss rate contained the true value in only 40.3% to 85.7% of cases. Reducing the sample size by a design effect raised coverage to between 93.3% and 94.0%. A power study suggests recording at least ten minutes on each network once we move on to real measurements.**

***Index Terms*: packet loss, Gilbert-Elliott model, burstiness, goodness of fit, Bayesian estimation, design effect, Monte Carlo simulation**

I. Introduction

Most people have run into packet loss without calling it that. A video call on poor Wi-Fi freezes for a second, or a word drops out of the audio, and then the call picks up again. Underneath, the application is sending small packets at a fixed rate, and a few of them never reach the other end. The fraction that goes missing is called the loss rate. It is the figure written into service agreements, and protocol designers rely on it when they choose retransmission timers or decide how much error correction a stream should carry.

What the loss rate does not show is how the losses are spread out in time. If ten packets are lost at scattered points over a minute, the decoder can hide each gap and the user hardly notices. Ten losses in a row produce an obvious freeze. Textbook models usually sidestep this by assuming that every packet is lost independently with the same small probability p. The assumption is convenient: it makes the number of losses in a short window Binomial, and so close to Poisson, and the time between losses exponential. Once losses start to cluster, neither result is correct, and the error bars worked out from them are wrong too.

Our aim was to find out how badly independence fails, which standard tests pick up the failure, and what it does to the interval estimates usually reported for the loss rate. The work is split into two phases, and this report covers Phase 1, a simulation study. Because we set the true parameters of each network ourselves, we can compare every estimate and test with the correct answer before the same code is used on real data. Phase 2 will apply that code to real captures.

The report is organised as follows. Section II gives some background, Section III sets out the loss model and how the data were generated, Section IV describes the statistical methods and Section V reports the results. Section VI discusses what the results mean and where the study is limited, and Section VII concludes.

II. Background

Modelling bursty errors with a hidden two-state chain goes back to E. N. Gilbert, who used it for errors on telephone lines in 1960. In 1963 E. O. Elliott extended the model so that errors could occur in either state. The Gilbert-Elliott model that came out of this is still the usual starting point for bursty loss.

It has also been known for many years that Internet losses are not independent. Measurement studies in the 1990s showed that probe packets sent close together were often lost together, and that losses in TCP transfers came in correlated episodes. The timing of loss in multicast traces could be described reasonably well with Markov chains. Later studies fitted Gilbert models to voice traffic and related the length of a loss burst to how poor a call sounds.

Most of this earlier work sets out to describe the loss process. Our interest is more statistical. We want to know which standard tests detect the dependence, how much data they need before they do, and how far ordinary interval estimates drift from the truth when the dependence is ignored. For the last of these we use the design effect from survey sampling, where it corrects for people being sampled in clusters instead of one at a time.

III. Loss Model and Data

A. The Gilbert-Elliott Channel

We treat the network as being in one of two hidden states, Good (G) or Bad (B). We cannot observe the state itself, only what it does to the packets. Before each packet, the channel switches from Good to Bad with probability p and from Bad back to Good with probability r. In the Good state a packet is lost with a small probability h_G; in the Bad state the probability h_B is much larger. Over a long run, the flow

TABLE I
SCENARIO PARAMETERS AND THE TRUE LOSS RATE THEY IMPLY

Network	p	r	h_G	h_B	π_B	Bad spell	True loss
Wired	0.0005	0.3	0.002	0.5	0.17%	3.3	0.283%
Wi-Fi	0.002	0.15	0.004	0.6	1.32%	6.7	1.184%
Mobile	0.004	0.08	0.006	0.7	4.76%	12.5	3.905%

Bad spell is the average stay in the Bad state, $1/r$ packets. True loss comes from (1).

into the Bad state has to equal the flow out of it. This gives the share of time spent in Bad, and from it the overall loss rate:

$$\pi_B = \frac{p}{p+r}, \qquad P(\text{loss}) = (1-\pi_B)\,h_G + \pi_B\,h_B. \tag{1}$$

The number of packets spent in a state before leaving it is geometric, with a mean of $1/p$ packets for Good and $1/r$ for Bad. A burst of losses happens whenever the channel drops into the Bad state for a few packets.

B. Scenarios and Sampling

Table I lists the parameters for the three networks. We chose them to give a low, a medium and a high loss rate, with longer Bad spells on the worse networks. Every network sends a constant 50 packets per second, one every 20 ms, which is close to the rate of a voice call. Each run lasts 3,600 s, so there are 180,000 packets per network and 540,000 overall. All random numbers come from the seed 2026, so anyone rerunning the code gets exactly the numbers in this report.

C. Simulation

The chain could be simulated by stepping forward one packet at a time, but that is slow. Instead, we draw the length of each stay in Good or Bad directly from the geometric distribution and place the stays one after another. Then, for every packet, we decide whether it is lost using the loss probability of the state it falls in. The process is the same, just far quicker to generate. We take four things from each trace: the number of losses in each 1-s window (3,600 counts per network), the gaps between successive losses, the lengths of bursts (runs of consecutive losses), and how often a delivered (N) or lost (L) packet is followed by each type of packet. Fig. 1 plots the first few minutes of each trace. On Wi-Fi and the mobile hotspot the clumping can already be seen by eye.

IV. METHODOLOGY

A. Fitting the Distributions

For the per-window loss counts $x_1, \dots, x_n$, the first model we fit is the Poisson distribution, whose maximum-likelihood estimate is simply the mean count, $\hat{\lambda} = \bar{x}$. The natural alternative is the Negative Binomial:

$$P(X=k) = \binom{k + r_{NB} - 1}{k} q^{r_{NB}} (1-q)^k, \tag{2}$$

$$\operatorname{Var}(X) = \mu + \mu^2 / r_{NB}. \tag{3}$$

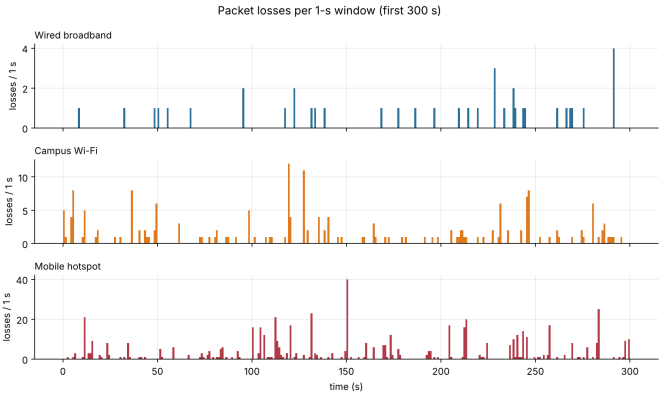


Fig. 1. Losses per second during the first five minutes of each simulated trace. On Wi-Fi and the mobile hotspot the losses come in clusters instead of one at a time.

The Negative Binomial can be thought of as a Poisson whose rate changes over time, which is what a network moving between good and bad periods produces. The extra term μ^2/r_{NB} lets the variance exceed the mean, something the Poisson cannot do. We fit it by fixing the mean at $\bar{x}$ and searching for the r_{NB} that maximises the likelihood.

For the gaps between losses we begin with the exponential distribution, with $\hat{\lambda} = 1/\bar{t}$, and compare it with a mixture of two exponentials known as the two-phase hyperexponential:

$$f(t) = w\lambda_1 e^{-\lambda_1 t} + (1-w)\lambda_2 e^{-\lambda_2 t}. \tag{4}$$

The idea behind the mixture is that gaps come in two types. Short gaps, with the fast rate λ_1, occur within a burst. Long gaps, with the slow rate λ_2, occur between bursts. We never know which type a particular gap is, so we fit the mixture with the Expectation-Maximisation (EM) algorithm. In each iteration the E-step works out the probability that each gap is a short one, and the M-step updates w, λ_1 and λ_2 with these probabilities as weights. We stop once the fit no longer improves. Models are compared with the Akaike Information Criterion, $\text{AIC} = 2k - 2\ln L$. It rewards a good fit but adds a penalty for each extra parameter, so a lower value means a better model.

B. Goodness of Fit

We check the count models by comparing the observed number of windows with each count against the number the model expects, using the Pearson statistic $\chi^2 = \sum_i (O_i - E_i)^2/E_i$. The test is unreliable when expected counts in the tail are small, so tail bins are merged until every expected count is at least 5. The degrees of freedom are the number of bins, minus one, minus the number of fitted parameters. We also apply the index-of-dispersion test. If the counts are Poisson, $(n-1)s^2/\bar{x}$ follows a χ^2_{n-1} distribution, so a variance well above the mean gives a very large value of the statistic.

For the gaps we use the Kolmogorov-Smirnov (KS) distance $D = \sup_t |F_n(t) - F(t)|$, the largest vertical distance between the cumulative curve of the data and that of the model. One

complication is that the standard KS tables assume the model was fixed before the data were seen. Here the rate is estimated from the same data, which makes the fit look better than it is and leaves the textbook p-value too large. We get the p-value from a parametric bootstrap instead. A synthetic data set is simulated from the fitted model, the model is refitted, and the new D is recorded. Doing this 2,000 times for the exponential and 500 times for the mixture shows the range of D to expect if the model were in fact correct.

C. Testing Independence

The Wald-Wolfowitz runs test works on the 0/1 sequence of delivered and lost packets and counts the number of runs R, a run being a stretch of identical outcomes. With n_1 losses and n_0 deliveries, independence gives

$$E[R] = \frac{2n_1n_0}{n} + 1, \quad \mathrm{Var}(R) = \frac{2n_1n_0(2n_1n_0 - n)}{n^2(n-1)}. \quad (5)$$

When losses tend to occur together, the sequence switches between lost and delivered less often than it would by chance, and R ends up smaller than expected.

The Markov-chain test asks the question more directly: does the outcome for one packet depend on the packet before it? Under H_0 all packets share a single loss probability. Under H_1 there are two, $a = P(L \mid N)$ after a delivered packet and $b = P(L \mid L)$ after a lost one. Writing n_{xy} for the number of times outcome x is followed by outcome y,

$$\hat{a} = \frac{n_{NL}}{n_{NN} + n_{NL}}, \qquad \hat{b} = \frac{n_{LL}}{n_{LN} + n_{LL}}, \quad (6)$$

$$G^2 = 2(\ln L_1 - \ln L_0). \quad (7)$$

Because H_1 has one parameter more than H_0, Wilks' theorem says that G^2 is approximately χ^2_1 when H_0 holds. The two estimates are useful on their own as well. The ratio b/a measures how much a loss raises the chance of the next loss, and $1/(1-b)$ gives the mean burst length.

The three networks are compared with rank-based tests, since the counts are mostly zero, strongly skewed and far from normal. The Kruskal-Wallis test compares all three networks together. Pairwise Mann-Whitney tests compare them two at a time, with a Bonferroni correction so that running three tests does not push up the false-alarm rate. A χ^2 test of homogeneity on the totals of lost and delivered packets checks whether the loss proportion is the same on every network. All tests use $\alpha = 0.05$.

D. Bayesian Estimation

We estimate the loss rate itself the Bayesian way. The prior is a flat $\mathrm{Beta}(1, 1)$, which treats every loss rate between 0 and 1 as equally likely before any data are seen. After k losses in n packets the posterior becomes $\mathrm{Beta}(1 + k,\ 1 + n - k)$, so updating just means adding the observed losses and deliveries to the prior counts. We report the posterior mean with the 95% credible interval, and for Campus Wi-Fi we also show how the posterior narrows as more traffic is observed. To check whether the networks really are ordered by loss rate, we draw 200,000 values from each posterior and count the share of draws in which Wired $<$ Wi-Fi $<$ Mobile.

The burst parameters are handled the same way. We place separate posteriors on a and b, $a \sim \mathrm{Beta}(1 + n_{NL}, 1 + n_{NN})$ and $b \sim \mathrm{Beta}(1 + n_{LL}, 1 + n_{LN})$, and these give credible intervals for b/a and for the burst length. As a check on the independent model, we also use it to predict how often a window should contain at least 5 losses, and compare that with how often it actually happens.

E. Design Effect

The variance of a sum shows why bursts cause problems for interval estimates:

$$\mathrm{Var}\Big(\sum_i X_i\Big) = \sum_i \mathrm{Var}(X_i) + 2\sum_{i<j} \mathrm{Cov}(X_i, X_j). \quad (8)$$

For independent packets every covariance term is zero, and the variance of the loss count is just $np(1-p)$. Within a burst, though, neighbouring packets tend to have the same outcome. The covariances are positive and the true variance is larger. The standard interval ignores this and comes out too narrow.

To correct for it, we divide each trace into blocks of $m = 500$ packets (10 s) and compare the observed spread of the block counts with the spread independence would give:

$$\mathrm{deff} = \frac{s^2_{\mathrm{block}}}{m\,\hat{p}(1-\hat{p})}, \qquad n_{\mathrm{eff}} = \frac{n}{\mathrm{deff}}, \quad (9)$$

where deff is not allowed to fall below 1. You can read n_{eff} as the number of independent packets that carry the same information as our correlated ones. The adjusted posterior uses these effective counts, $\mathrm{Beta}(1 + k/\mathrm{deff},\ 1 + (n-k)/\mathrm{deff})$, and the interval it gives is wider and closer to the truth.

F. Monte Carlo Checks

Two of our questions cannot be answered from a single trace, so we answer them by repeating the experiment many times. The first is power, the probability that each test notices the bursts. For capture lengths between 15 s and 600 s, we simulate 2,000 captures per network and record how often each test rejects independence. The second is coverage: how often a 95% interval actually includes the true loss rate. For this we simulate 2,000 captures of 10 minutes for each network, build the naive and the adjusted interval for every capture, and count how many contain the true value. With 2,000 repetitions, each of these proportions has a standard error of about 0.011.

V. Results

A. Loss Counts

None of the three networks fits the Poisson model, and Table II shows why. A Poisson distribution has its variance equal to its mean. The observed variance is 1.59, 5.31, 10.17 times the mean, and the dispersion test gives $p < 10^{-10}$ in each case. The Negative Binomial fits much better and has a lower AIC on all three networks. The largest gain is on Mobile, where the AIC falls from 25,042 to 11,700.

Most of the difference between the two models lies in the tail. For Wi-Fi, the Poisson model expects about 1.98 seconds

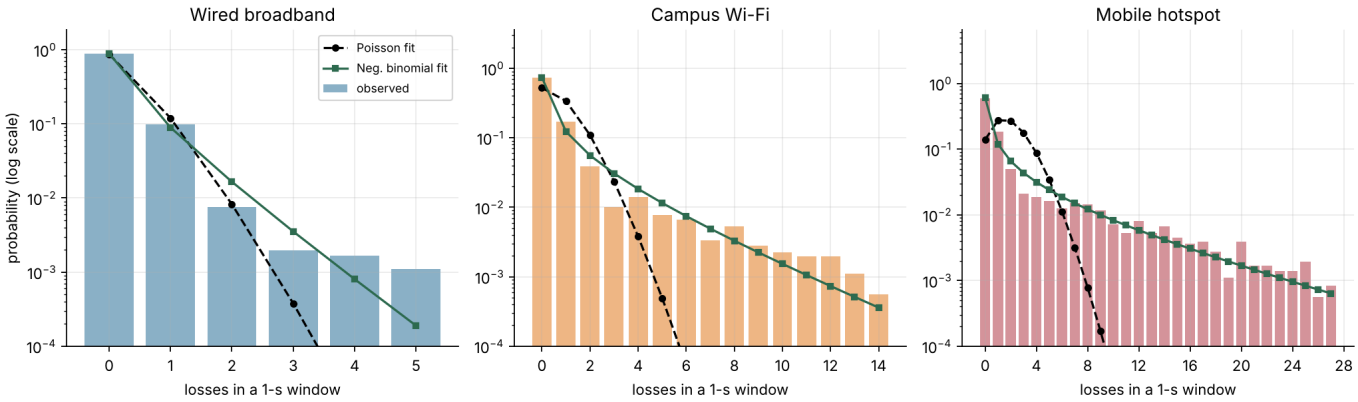


Fig. 2. Probability of k losses in one second, on a log scale. The Poisson fit (dashed) falls well below the observed tail, while the Negative Binomial (green) stays close to it.

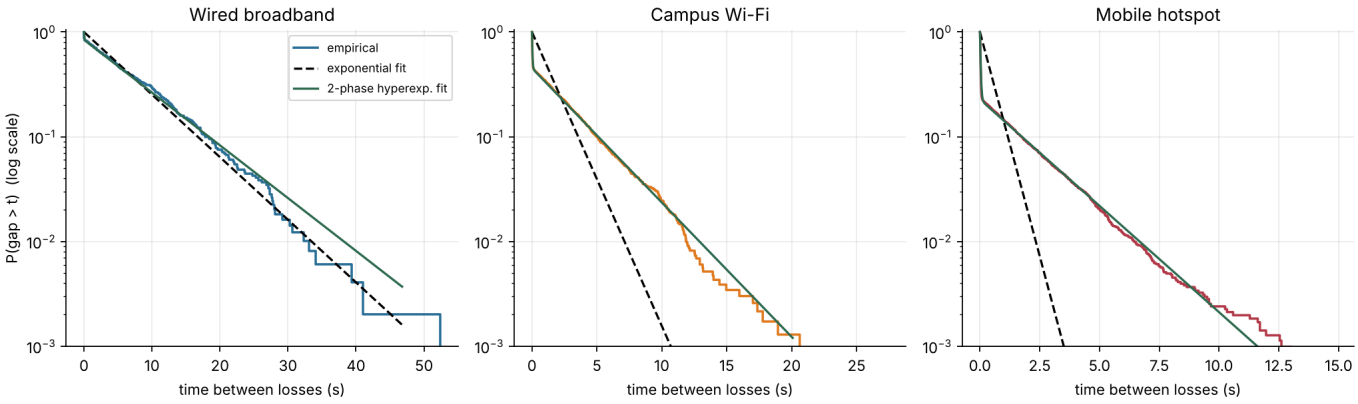


Fig. 3. Probability that the gap between two losses is longer than t, on a log scale. The exponential fit is a straight line (dashed); the two-phase mixture (green) bends with the data.

TABLE II
LOSSES PER SECOND AND HOW WELL THE COUNT MODELS FIT

Network	Loss rate	Mean / s	Variance	D	χ^2 Poisson	AIC Poisson	AIC Neg. Bin.	Seconds with $\geq$5 losses
Wired	0.275%	0.138	0.219	1.59	22.5	3,143	2,992	7 / 0.00132 / 0.9
Wi-Fi	1.292%	0.646	3.430	5.31	2,544.2	10,607	7,186	131 / 1.98 / 124.2
Mobile	3.937%	1.969	20.017	10.17	15,840.8	25,042	11,700	487 / 180 / 492.0

D is the variance divided by the mean. The last column gives the number of seconds with at least 5 losses: observed / expected under Poisson / expected under Negative Binomial. Every Poisson test gives $p < 10^{-10}$.

with 5 or more losses in an hour. The trace contains 131, and the Negative Binomial expects 124.2. Fig. 2 shows this on a log scale: the Poisson curve drops away almost at once, while the observed counts and the Negative Binomial carry on further out.

B. Gaps Between Losses

The exponential model is rejected as well, with the bootstrap KS test failing it on every network (Table III). Fig. 3 makes the problem easy to see. On a log scale an exponential survival curve is a straight line, and the observed curves are not straight. They fall steeply at first, because many of the gaps are very short, and then level off, because the remaining gaps are long.

The two-phase mixture reproduces this shape well. The mean gap in its fast phase is 29 to 33 ms, one to two packets, which matches gaps inside a burst. The slow phase has a mean of 2.1 to 8.6 s and matches the quiet periods between bursts. The share of gaps in the fast phase rises from 16% on Wired to 77% on Mobile, in line with more of the mobile link's losses falling inside bursts. The mixture passes the KS test on Wired ($p = 0.154$) and lowers the AIC considerably on all three networks. It is still rejected on Wi-Fi and Mobile, so it is a large improvement on the exponential but not a perfect description of the gaps.

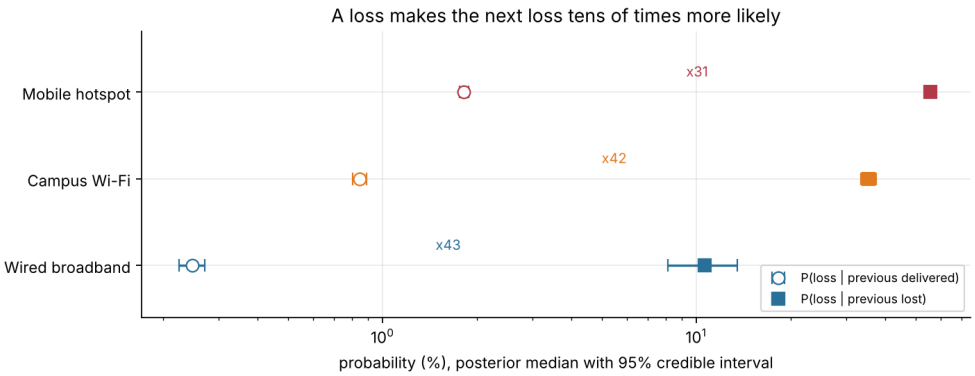


Fig. 4. Posterior medians and 95% credible intervals for $P(L \mid N)$ (open) and $P(L \mid L)$ (filled). On all three networks a loss sharply raises the chance that the next packet is lost too.

TABLE III
GAPS BETWEEN LOSSES: EXPONENTIAL AND TWO-PHASE FITS

Net.	w	Fast (ms)	Slow (s)	KS p exp.	KS p mix.	AIC exp.	AIC mix.
Wired	16%	32	8.6	< 0.0005	0.154	2,950	2,644
Wi-Fi	55%	33	3.4	< 0.0005	< 0.002	6,682	1,368
Mobile	77%	29	2.1	< 0.0005	< 0.002	4,563	-15,477

w is the share of gaps in the fast phase. Fast and slow are the average gap in each phase.

TABLE IV
TESTS OF INDEPENDENCE AND BURST PARAMETERS

Net.	Runs	$E[R]$	z	G^2	$P(L\|N)$	$P(L\|L)$	b/a	Burst
Wired	887	988	-43.5	288.4	0.25%	10.5%	43	1.12
Wi-Fi	3,005	4,593	-146.7	4,498.6	0.85%	35.4%	42	1.55
Mobile	6,293	13,617	-228.2	18,560.2	1.82%	55.6%	31	2.25

b/a and burst length (in packets) are posterior medians. All $p < 10^{-10}$.

C. Do Losses Depend on Each Other?

They clearly do. Both independence tests reject H_0 with $p < 10^{-10}$ on every network (Table IV). For Wi-Fi we count 3,005 runs, against about 4,593 expected under independence, a z-score of -146.7. The Markov estimates make the effect easier to picture. On Wi-Fi a packet is lost 0.85% of the time when the packet before it was delivered, but 35.4% of the time when the packet before it was lost.

The posterior medians of b/a are 43, 42, 31 for Wired, Wi-Fi and Mobile, so a single loss makes the next loss 31 to 43 times more likely (Fig. 4). The posterior probability that $b > a$ is 1.000 on every network, so the direction of the effect is not in doubt. The average bursts are short, at 1.12, 1.55, 2.25 packets, but they happen far more often than they would by chance.

D. Comparing the Networks

The three networks differ clearly, as expected from how they were set up. The Kruskal-Wallis statistic is $H = 937.3$, the χ^2 homogeneity statistic is 7,143.0 on 2 degrees of freedom, and even the largest Bonferroni-corrected pairwise p-value is $< 10^{-10}$. The posterior mean loss rates are 0.276%, 1.293%, 3.938% (Table V), and the posterior probability of the ordering Wired $<$ Wi-Fi $<$ Mobile is 1.000.

Fig. 5 shows the Bayesian estimates. With only 10 s of Campus Wi-Fi traffic the estimate is 3.78% [2.30%, 5.62%], a wide interval. After 3,600 s it has narrowed to 1.29% [1.24%, 1.35%].

The predictive check shows how poorly the independent model handles seconds with many losses. For Wi-Fi it puts the probability of a second with at least 5 losses at 0.00047. In the trace such seconds make up 0.036 of the total, about 77 times as often as predicted.

E. How Often Do the Intervals Work?

For us this is the key result. Across 2,000 simulated 10-minute captures, the naive 95% interval included the true loss rate in only 85.7%, 60.9%, 40.3% of cases (Table V, Fig. 6). On the burstiest network it missed the true value more often than it caught it.

The design effects are 1.52, 6.43 and 12.14. For Wi-Fi, this means our 180,000 packets hold roughly the same information as 28,000 independent packets would. With these smaller effective sample sizes, coverage returns to 93.3%, 94.0%, 94.0%, close to the intended 95%. The intervals do get wider; on Wi-Fi the average width grows from 0.245 to 0.563 percentage points. We think that is worth it for intervals that contain the true value about as often as they claim to.

The full Wi-Fi trace is a good example. Its true loss rate is 1.184%. The naive interval of [1.241%, 1.345%] lies entirely above this value, while the adjusted interval of [1.167%, 1.431%] contains it.

F. How Long Should a Capture Be?

Fig. 7 shows how the power of each test increases with capture length, and Table VI lists the shortest length we tried at which power reaches 0.8. With the runs test, the mobile network needs only 15 s and Wi-Fi 30 s. Wired needs 300 s, since its bursts are infrequent and short and evidence takes longer to build up. The runs test was the most powerful of the

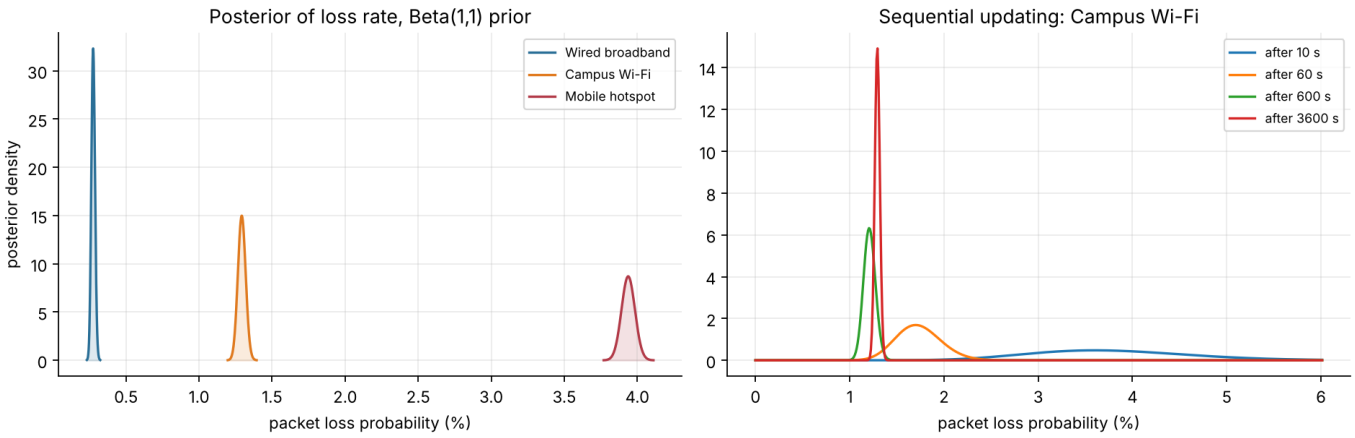


Fig. 5. Left: posterior distributions of the loss rate for the three networks. Right: the Wi-Fi posterior after 10 s, 60 s, 600 s and 3,600 s of traffic.

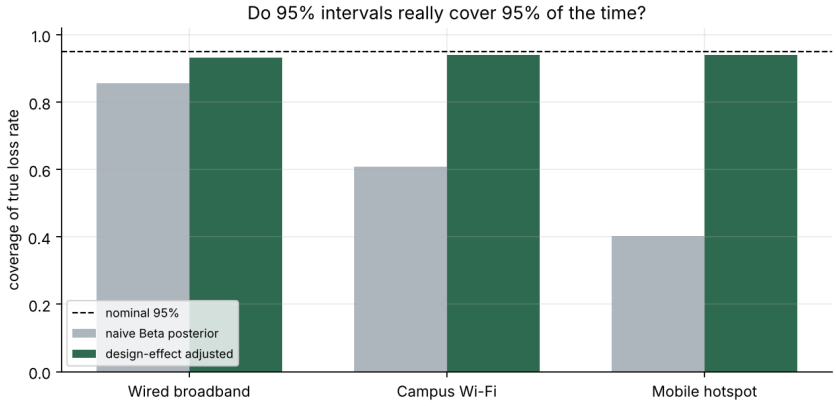


Fig. 6. Share of nominal 95% intervals that contain the true loss rate, before and after the design-effect correction.

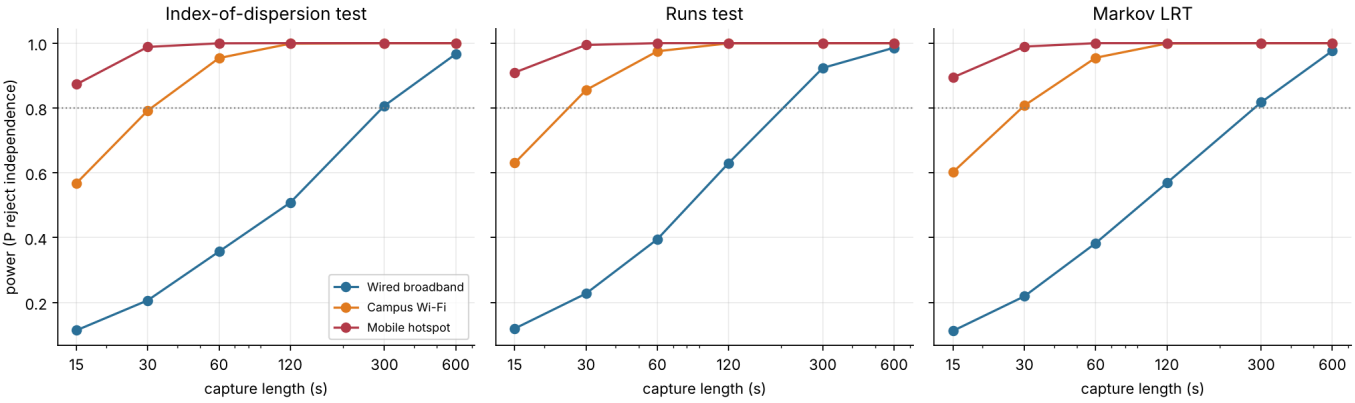


Fig. 7. Power to detect dependence as capture length grows, from 2,000 simulated captures per point. The dotted line marks a power of 0.8.

three on every network. At 600 s every test has a power of at least 0.97 on every network, which is why we plan to record at least ten minutes per network in Phase 2.

VI. Discussion and Limitations

It takes surprisingly little for independence to fail. A Bad spell of only a few packets is enough to break every model built on that assumption, and the loss rate alone gives no hint of it. Mobile loses roughly 14 times as many packets as Wired, yet on both networks a loss makes another loss tens of times more likely. Burstiness is a separate property of a link and has to be measured separately.

The coverage results show the problem is bigger than

TABLE V
LOSS-RATE ESTIMATES AND COVERAGE OF 95% INTERVALS

Network	Posterior mean	Naive 95% interval	deff	Adjusted 95% interval	Naive coverage	Adjusted coverage
Wired	0.276%	[0.252%, 0.300%]	1.52	[0.247%, 0.307%]	85.7%	93.3%
Wi-Fi	1.293%	[1.241%, 1.345%]	6.43	[1.167%, 1.431%]	60.9%	94.0%
Mobile	3.938%	[3.848%, 4.028%]	12.14	[3.636%, 4.263%]	40.3%	94.0%

Intervals are from the full 1-hour trace. Coverage is from 2,000 simulated 10-minute captures.

TABLE VI
SHORTEST CAPTURE (S) WITH POWER OF AT LEAST 0.8

Network	Dispersion	Runs	Markov	Min. power at 600 s
Wired	300	300	300	0.97
Wi-Fi	60	30	30	1.00
Mobile	15	15	15	1.00

picking the right model. A naive interval counts every packet as a new and independent piece of evidence. When packets within a burst share the same outcome, there is much less independent evidence than the packet count suggests, and the interval turns out too narrow. The effect grows as the network becomes burstier, which is why Mobile has the lowest naive coverage. The design-effect correction is simple, needs only block counts, and brings coverage close to 95% on all three networks.

There are limits to what we can claim. The main one is that the data are simulated, and from the same kind of model that motivated the study. The tests do not use this, and the true parameters are only used to judge the estimates, but real links have other effects, such as busy hours, that our model leaves out. Phase 2 is meant to deal with this.

There are also a few smaller points. The dispersion test compares the counts with a Poisson distribution, but counts out of 50 packets are really Binomial, with a variance slightly below the mean. This makes the test a little conservative, so it is more likely to understate burstiness than to overstate it. The runs and Markov tests rely on large-sample approximations. In very short captures their false-alarm rate may drift from 5%, although for hour-long traces it stays close to 5%. The Negative Binomial and the two-phase mixture are both still rejected on Wi-Fi and Mobile, so neither gives a complete description. The design effect depends on the block length we choose, and our 10-s blocks are much longer than any Bad spell, which should be enough. Finally, the headline figures come from one seeded trace per network, although the Monte Carlo checks repeat the whole experiment thousands of times.

VII. CONCLUSION AND FUTURE WORK

We tested whether packet losses on three simulated networks behave independently, and found that they do not. The Poisson and exponential models were rejected in every case. The per-second loss counts were 1.59, 5.31, 10.17 times more spread out than a Poisson model allows, the gaps split into short gaps within bursts and long quiet gaps between them, and each loss made the next one 31 to 43 times more likely. Most importantly, the standard 95% interval for the loss rate was right in only 40.3% to 85.7% of cases. Dividing the sample size by the design effect raised this to between 93.3% and 94.0%.

The next step is Phase 2. We will record steady streams of 50 packets per second on real wired, Wi-Fi and mobile links, for at least ten minutes each, and run the same code on them. After that we would like to estimate the Gilbert-Elliott parameters directly from the real traces using the Baum-Welch algorithm, try richer models for the tail, use the Phase 1 results as priors in a hierarchical Bayesian model across the networks, and compare loss at peak and off-peak hours.

REPRODUCIBILITY

Every result in this report was produced by one Python pipeline, built on NumPy, SciPy and Matplotlib and run with seed 2026. Running it again regenerates all of the tables and figures.